

Contact the AI Office of Ireland service desk

Direct URL: <https://forms.gov.ie/en/6a6ca2312b559f2106415768>

Response ID	9eb484b6-08d9-4182-97b1-edb0985d00e6
Time submitted	25/09/2026 at 4:32 pm

Your details

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7. Does your query:	Involve a concern relating to an AI system under the EU AI Act

Check if we can handle your complaint under the EU AI Act

8. Which part of the Act is your concern about?	I am not sure
9. What is the name of the AI system and/or operator forming the subject of your concern?	See Git Hub repo https://github.com/BorealisCodes/NDH-Research-Pilot/tree/df1f30e88e4fdd23c2e162d3bd69c0b46cd5f559/climate-studies https://github.com/BorealisCodes/Trauma-Informed-Systems-Design/tree/main/docs/Microsoft
10. Please describe your concern.	Energy Profiles of Large Scale AI Systems: A Mathematical and Rights Based Assessment for Irish Regulatory Authorities Executive Summary This report provides a mathematical comparison between brute force black box AI computation and spectral, structure aware computational methods, framed within Ireland's obligations under the European Convention on Human Rights (ECHR). Recent ECtHR jurisprudence (2024) affirms that climate protection is a human rights obligation under Articles 2 and 8. High energy AI systems therefore fall within regulatory scope. Spectral and manifold aware computational approaches—such as Riemann Minimal Entropy Collapse and Compressed Manifold Reachability—offer substantially lower energy profiles and reduced environmental impact compared to brute force methods. --- 1. Climate Rights Under the ECHR 1.1 Article 2 – Right to Life States must prevent foreseeable environmental harm that poses risks to life. 1.2 Article 8 – Right to Private and Family Life Environmental degradation, including climate impacts, can violate Article 8 when authorities fail to regulate high risk activities. 1.3 Relevant Judgments (2024) - Verein KlimaSeniorinnen Schweiz and Others v. Switzerland The Court held that inadequate climate mitigation measures breached Article 8. - Duarte Agostinho and Others v. Portugal and 32 Other States

11. If you can, tell us what you believe the AI system or organisation has done wrong, or failed to do, with regard to the EU AI Act?

The Court recognized cross-border climate impacts as falling under Convention obligations.

2. Mathematical Energy Profiles of AI Systems

2.1 Brute Force Black Box AI

2.1.1 Computational Model

Brute force AI relies on dense matrix multiplication:

$$\text{Compute Load} \approx O(n^3)$$

Energy consumption scales with FLOPs:

$$E_{\text{brute}} = \alpha \cdot \text{FLOPs}$$

2.1.2 Thermal and Water Load

Cooling demand:

$$Q_{\text{cool}} = \beta \cdot P_{\text{heat}}$$

Water usage:

$$W = \gamma \cdot Q_{\text{cool}}$$

2.1.3 Grid Stress

Peak load:

$$P_{\text{peak}} = \sum_{i=1}^k P_i$$

`yaml

cff-version: 1.2.0

message: "Please cite this research as follows."

title: "Riemann Minimal Entropy Collapse - A Topological Proof by Contradiction of Non-Trivial Zeros via Spectral Rigidity in Lean 4 & Comparator"

version: "1.0.0"

doi: "10.5281/zenodo.22730476"

date-released: "2026-09-13"

authors:

- name: "Jonathan f(n)"

Reference B – Compressed Manifold Reachability

`yaml

cff-version: 1.2.0

message: "Please cite this research as follows."

title: "Compressed Manifold Reachability - A Formally Proven Polynomial-Scaling Safety Algorithm for High-Dimensional Robotic Systems ($d \geq 6$)"

version: "1.0.0"

doi: "10.5281/zenodo.22683309"

date-released: "2026-09-10"

authors:

- name: "Jonathan f(n)"

12. How has this affected you or others?

3. Comparison Tables

Table 1 – Computational Complexity

Method	Complexity	Notes
Brute Force AI	$O(n^3)$	Dense matrix multiplication; high FLOPs
Spectral Methods	$O(n \log n)$	Dominant eigenmodes; compressed manifold representations

Table 2 – Energy Consumption

Method	Energy Use (Relative)	Drivers
Brute Force AI	Very High	FLOPs, GPU clusters, cooling
Spectral Methods	Low	Reduced FLOPs, manifold compression

	--- Table 3 — Water Usage Method Water Demand Notes ----- ----- ----- Brute? ForceAI High Cooling towers, evaporative cooling Spectral Methods Minimal Lower thermal output --- Table 4 — Grid Stress Method Peak Load Notes ----- ----- ----- Brute? ForceAI High Multi? MWdata centers Spectral Methods Low Smaller clusters, lower heat --- Table 5 — Rights? BasedRisk (ECHR) Method Article 2 Risk Article 8 Risk Notes ----- ----- ----- ----- Brute? ForceAI Elevated Elevated Foreseeable environmental harm Spectral Methods Low Low Aligns with mitigation obligations Reference A — Riemann Minimal Entropy Collapse
13. Did you contact the organisation involved?	Yes
14. Did you receive a response?	Yes
15. Please upload the correspondence	Gmail - Fw_ Follow? Up to Case TrackingID#2607230040012918 — Request for Correct Routing and HRD? AlignedSupport(1).PDF
16. Have you already raised this issue elsewhere?	Yes, an Irish authority
17. Please share the name of the MSA and which country it is based in (if applicable)	https://github.com/BorealisCodes/NDH-Research-Pilot/blob/main/climate-studies/ai-ethics-climate-charities-and-regulators.txt

Supporting documents

18. Would you like to include some supporting documents?	Yes
19. Please upload your supporting document here	Omnibus Bash Script .pdf
20. Do you have another supporting document to add?	Yes
21. Please upload your supporting document here	? The Jargon Decoupling Report- An Epistemic Post-Mortem (1).pdf

Declaration

22. I confirm that the information provided is accurate to the best of my knowledge.	Yes
23. I understand that my submissions may be shared with a relevant authority for assessment.	Yes

This is for your personal records only and is not an official receipt.